- The temperature of the engine fins and the temperature of the cooling air

Advantages and disadvantages of air cooling system:

Advantages:

- In this type due to the absence of radiator fan and pumping unit, engine weight is considerably reduced
- Due to the absence of radiator arrangement, rust and deposits formation are avoided
- Engine occupies only less space
- As the engine needs no water jacket, design of the engine is simple
- The system need no water filling in the radiator
- The engine can be operated at all operating conditions (cold and hot regions)
- Freezing and evaporation of water do not occur as in case of water cooling system

Disadvantages:

- Air cool system is not suitable for CSMO cylinder engines
- As this type of engine is cooled by air, the efficiency of engine decreases
- More noise is produced
- Heat transfer rate is lower with air cooling as compared to water cooling
- Blower arrangement is needed for bigger size engines

7.2 WATER COOLING SYSTEM

In this method, the water is circulated inside the water jackets of the engine cylinder block and the cylinder head and rejects the heat in the cylinder block and head. As the water absorbs the heat from the cylinder it gets vapourized. It may cause insufficient water to be present in the cooling line. Hence excess water must be added to overcome the vapourization loss. In order to overcome this difficulty radiator arrangement is used to cool the hot water coming out from the engine water jackets. There are three types of water cooling systems followed in engines which are,

1. Direct (or) non return cooling system
2. Thermosyphon or natural circulation cooling system

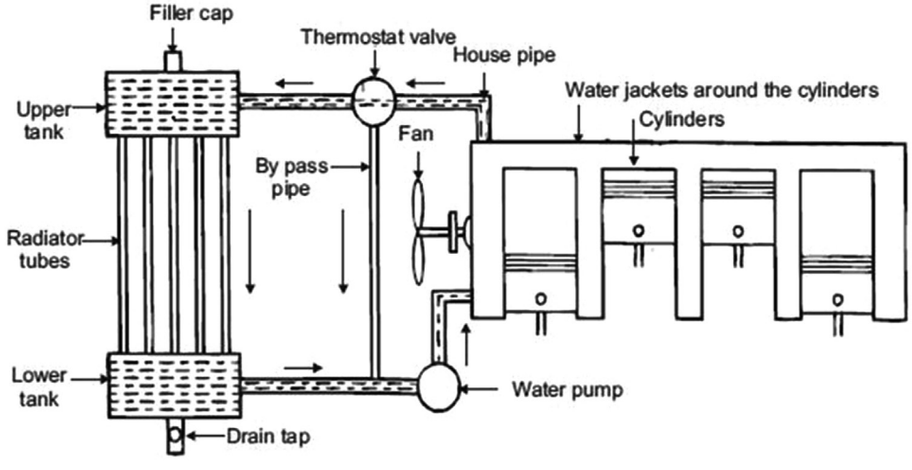


Figure 7.2.1 Water cooling system

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